STAFF REVIEW · H446 277 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. An attribute depends on key through another non-key attribute.

2. Search speed on indexed fields.

3. Atomic fields and no repeating groups.

4. $A \wedge (B \vee \neg B) = A \wedge 1 = A$.

I Do — source worked steps

1. Predict first — Store customer name once in Customer and reference by CustomerID in Order.

2. Model answer — Explain the reasoning and justify the output.

We Do — guided CHECK

1. Predict first
ProductName depends on ProductID rather than the whole composite sales-line identifier, so separate it.
Model answer
Explain the reasoning and justify the output.

You Do — source mark schemes

1. 1NF atomic/no repeating groups [2]; 2NF removes partial dependencies on composite key [2]; 3NF removes problematic transitive dependencies [2].

2. CustomerName repeated for one CustomerID [2]; Customer(CustomerID PK, CustomerName) [2]; Order(OrderID PK,CustomerID FK) [2].

3. Index helps locate selected names [2], can reduce scanning [1]; index uses storage [1], maintenance on writes [1], duplicates may need extra row matching [1].

Exit Ticket — source answer

1. False. Normalisation and indexing have different purposes.